CSA3714-SOFTWARE TESTING

LAB EXPERIMENTS

EXPERIMENT NO: 1

TOPIC: ATM SYSTEM – BUG IDENTIFICATION

OBJECTIVE

To identify and document software defects in an ATM system.

FEATURES TO BE TESTED

1. Card validation
2. User authentication
3. Account type selection
4. Cash withdrawal
5. Transaction and daily limits
6. Money denominations
7. Receipt printing
8. PIN change

BUG REPORT

Bug ID	**Bug Name**
ATM001	Invalid Card
ATM002	Invalid PIN
ATM003	Invalid Account Type
ATM004	Insufficient Balance
ATM005	Transaction Limit
ATM006	Daily Limit
ATM007	Invalid Money Denominations
ATM008	Receipt Not Printed
ATM009	PIN Change Mismatch
ATM010	Cash Not Dispensed

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TEST CASES

Test Case 1: Invalid Card

Test Case ID: ATM001

Steps: Insert an invalid ATM card.

Expected Result: The ATM should reject the card and display an error message.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 2: Invalid PIN

Test Case ID: ATM002

Steps: Insert a valid card and enter an incorrect PIN.

Expected Result: An invalid PIN message should be displayed.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 3: Invalid Account Type

Test Case ID: ATM003

Steps: Select an unsupported account type.

Expected Result: The ATM should reject the selection and display an error message.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 4: Insufficient Balance

Test Case ID: ATM004

Steps: Enter a withdrawal amount greater than the available balance.

Expected Result: Withdrawal should be rejected.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 5: Transaction Limit

Test Case ID: ATM005

Steps: Enter an amount exceeding the permitted transaction limit.

Expected Result: The ATM should display a transaction limit error.

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Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 6: Daily Limit

Test Case ID: ATM006

Steps: Attempt to withdraw beyond the daily withdrawal limit.

Expected Result: The transaction should be rejected.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 7: Invalid Money Denominations

Test Case ID: ATM007

Steps: Enter an amount that cannot be dispensed using the available notes.

Expected Result: The ATM should reject the amount and display a suitable message.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 8: Receipt Not Printed

Test Case ID: ATM008

Steps: Complete a withdrawal and request a receipt.

Expected Result: The ATM should print the receipt or display a printer error.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 9: PIN Change Mismatch

Test Case ID: ATM009

Steps: Enter different values in the new PIN and confirm PIN fields.

Expected Result: The ATM should reject the PIN change and display a mismatch message.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 10: Cash Not Dispensed

Test Case ID: ATM010

Steps: Perform a valid cash withdrawal and check the cash dispenser and account balance.

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Expected Result: Cash should be dispensed, and the account should be debited correctly.

Actual Result: To be observed during testing.

Status: Pass/Fail

RESULT

Thus, the ATM system was examined, and possible software defects were identified and documented.

EXPERIMENT NO: 2

TOPIC: ARMS PORTAL – POSITIVE TESTING

OBJECTIVE

To test the basic functions of the ARMS Portal using valid inputs.

FEATURES TO BE TESTED

1. User login
2. Student profile
3. Academic records
4. Student information update
5. Attendance details
6. Marks display
7. Logout

BUG REPORT / TEST CASE ID TABLE

Bug ID	**Bug Name**
ARMS001	Valid Login
ARMS002	Student Details Not Displayed
ARMS003	Academic Records Missing
ARMS004	Student Details Not Updated
ARMS005	Attendance Display Error
ARMS006	Marks Display Error
ARMS007	Logout Failure

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TEST CASES

Test Case 1: Valid Login

Test Case ID: ARMS001

Steps: Open the ARMS Portal, enter valid login credentials, and click Login.

Expected Result: The user should log in successfully, and the dashboard should be displayed.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 2: View Student Details

Test Case ID: ARMS002

Steps: Log in and open the student profile.

Expected Result: Student details should be displayed correctly.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 3: View Academic Records

Test Case ID: ARMS003

Steps: Open the academic records section.

Expected Result: Academic records should be displayed correctly.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 4: Update Student Details

Test Case ID: ARMS004

Steps: Enter valid information in an editable field and click Save.

Expected Result: The updated information should be saved successfully.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 5: View Attendance

Test Case ID: ARMS005

Steps: Open the attendance section.

Expected Result: Attendance details should be displayed correctly.

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Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 6: View Marks

Test Case ID: ARMS006

Steps: Open the marks section.

Expected Result: Marks should be displayed correctly.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 7: Logout

Test Case ID: ARMS007

Steps: Log in and click the Logout button.

Expected Result: The user should be logged out and redirected to the login page.

Actual Result: To be observed during testing.

Status: Pass/Fail

RESULT

Thus, positive test cases were prepared to verify the basic functions of the ARMS Portal using valid inputs.

EXPERIMENT NO: 3

TOPIC: ARMS PORTAL – NEGATIVE TESTING

OBJECTIVE

To test the ARMS Portal using invalid inputs and verify its error-handling functions.

FEATURES TO BE TESTED

1. Login validation
2. Empty field validation
3. Student detail validation
4. Unauthorized access
5. Search validation
6. Session management
7. Logout

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BUG REPORT

Bug ID	Bug Name
ARMS-N001	Invalid Login Credentials
ARMS-N002	Empty Fields Accepted
ARMS-N003	Invalid Student Details Accepted
ARMS-N004	Unauthorized Access Allowed
ARMS-N005	Invalid Search Results
ARMS-N006	Session Timeout Error
ARMS-N007	Logout Failure

TEST CASES

Test Case 1: Invalid Login Credentials

Test Case ID: ARMS-N001

Steps: Enter an incorrect username or password and click Login.

Expected Result: Login should be rejected, and an error message should be displayed.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 2: Empty Fields Accepted

Test Case ID: ARMS-N002

Steps: Leave the username and password fields empty and click Login.

Expected Result: The portal should display validation messages.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 3: Invalid Student Details Accepted

Test Case ID: ARMS-N003

Steps: Enter invalid data in an editable student information field and click Save.

Expected Result: Invalid data should be rejected with a suitable error message.

Actual Result: To be observed during testing.

Status: Pass/Fail

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Test Case 4: Unauthorized Access Allowed

Test Case ID: ARMS-N004

Steps: Try to access a restricted page without logging in.

Expected Result: Access should be denied, and the login page should be displayed.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 5: Invalid Search Results

Test Case ID: ARMS-N005

Steps: Search using invalid or unavailable student details.

Expected Result: A suitable message should be displayed without showing incorrect records.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 6: Session Timeout Error

Test Case ID: ARMS-N006

Steps: Leave the portal inactive until the configured session timeout period expires.

Expected Result: The session should expire securely, and the user should be asked to log in again.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 7: Logout Failure

Test Case ID: ARMS-N007

Steps: Log in, click Logout, and try to access a protected page using the previous session.

Expected Result: The previous session should be terminated, and access should be denied.

Actual Result: To be observed during testing.

Status: Pass/Fail

RESULT

Thus, negative test cases were prepared to verify whether the ARMS Portal handles invalid inputs and unauthorized access correctly.

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EXPERIMENT NO: 4

TOPIC: E-COMMERCE WEBSITE – POSITIVE TESTING

OBJECTIVE

To test the basic functions of an e-commerce website using valid inputs.

FEATURES TO BE TESTED

1. User registration
2. User login
3. Product search
4. Add to cart
5. Quantity update
6. Payment
7. Order placement

BUG REPORT / TEST CASE ID TABLE

Bug ID	**Bug Name**
EC001	User Registration Failure
EC002	Valid Login Failure
EC003	Product Search Error
EC004	Add to Cart Failure
EC005	Quantity Update Error
EC006	Payment Failure
EC007	Order Confirmation Missing

TEST CASES

Test Case 1: User Registration

Test Case ID: EC001

Steps: Enter valid registration details and click Register.

Expected Result: The account should be created successfully.

Actual Result: To be observed during testing.

Status: Pass/Fail

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Test Case 2: Valid Login

Test Case ID: EC002

Steps: Enter a registered username and correct password.

Expected Result: The user should log in successfully.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 3: Product Search

Test Case ID: EC003

Steps: Search for an available product using its name.

Expected Result: Relevant products should be displayed.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 4: Add to Cart

Test Case ID: EC004

Steps: Select an available product and click Add to Cart.

Expected Result: The selected product should be added to the cart.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 5: Quantity Update

Test Case ID: EC005

Steps: Increase the quantity of a product in the cart.

Expected Result: The quantity and total price should be updated correctly.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 6: Payment

Test Case ID: EC006

Steps: Complete the checkout using valid payment details.

Expected Result: Payment should be processed successfully, and the order should be recorded.

Actual Result: To be observed during testing.

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Status: Pass/Fail

Test Case 7: Order Confirmation

Test Case ID: EC007

Steps: Place an order successfully and view the order confirmation.

Expected Result: An order confirmation and order ID should be displayed.

Actual Result: To be observed during testing.

Status: Pass/Fail

RESULT

Thus, positive test cases were prepared to verify the basic functions of an e-commerce website using valid inputs.

EXPERIMENT NO: 5

TOPIC: E-COMMERCE WEBSITE – NEGATIVE TESTING

OBJECTIVE

To test an e-commerce website using invalid inputs and verify its error-handling functions.

FEATURES TO BE TESTED

1. Registration validation
2. Login validation
3. Product availability
4. Quantity validation
5. Cart validation
6. Payment validation
7. Address validation
8. Order processing

BUG REPORT

Bug ID	**Bug Name**
EC-N001	Invalid Registration Details
EC-N002	Invalid Login Credentials
EC-N003	Out-of-Stock Product Accepted
EC-N004	Invalid Quantity Accepted

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EC-N005	Empty Cart Checkout
EC-N006	Invalid Payment Accepted
EC-N007	Invalid Address Accepted
EC-N008	Incorrect Total Price
EC-N009	Duplicate Order
EC-N010	Order Confirmation Missing

TEST CASES

Test Case 1: Invalid Registration Details

Test Case ID: EC-N001

Steps: Enter an invalid email address during registration and click Register.

Expected Result: Registration should be rejected, and an error message should be displayed.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 2: Invalid Login Credentials

Test Case ID: EC-N002

Steps: Enter an incorrect username or password.

Expected Result: Login should be rejected.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 3: Out-of-Stock Product Accepted

Test Case ID: EC-N003

Steps: Try to add an out-of-stock product to the cart or place an order for it.

Expected Result: The website should prevent the purchase and display an out-of-stock message.

Actual Result: To be observed during testing.

Status: Pass/Fail

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Test Case 4: Invalid Quantity Accepted

Test Case ID: EC-N004

Steps: Enter zero or a negative quantity for a product.

Expected Result: The website should reject the invalid quantity.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 5: Empty Cart Checkout

Test Case ID: EC-N005

Steps: Try to proceed to checkout without adding any products to the cart.

Expected Result: Checkout should not be allowed.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 6: Invalid Payment Accepted

Test Case ID: EC-N006

Steps: Attempt payment using invalid payment details in a test environment.

Expected Result: Payment should be rejected, and an error message should be displayed.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 7: Invalid Address Accepted

Test Case ID: EC-N007

Steps: Enter an incomplete delivery address and try to place the order.

Expected Result: The website should display validation messages and prevent order placement.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 8: Incorrect Total Price

Test Case ID: EC-N008

Steps: Add products to the cart and compare the displayed total with the item prices and quantities.

Expected Result: The total price should be calculated correctly.

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Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 9: Duplicate Order

Test Case ID: EC-N009

Steps: Submit the same order repeatedly or refresh the order submission page.

Expected Result: Duplicate orders should not be created unintentionally.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 10: Order Confirmation Missing

Test Case ID: EC-N010

Steps: Complete an order and check whether confirmation is displayed.

Expected Result: A confirmation message and order ID should be displayed for a successful order.

Actual Result: To be observed during testing.

Status: Pass/Fail

RESULT

Thus, negative test cases were prepared to verify whether the e-commerce website handles invalid inputs, payment errors, and order-processing issues correctly.

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